Assessment of Skin Physiology Change and Safety After Intradermal Injections With Botulinum Toxin: A Randomized, Double-Blind, Placebo-Controlled, Split-Face Pilot Study in Rosacea Patients With Facial Erythema

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BACKGROUND Botulinum toxin (BTX) has been used cosmetically with good clinical efficacy and tolerable safety.

OBJECTIVE This randomized, double-blind, split-face clinical study aimed to investigate the efficacy and safety of intradermal BTX in patients with rosacea.

MATERIALS AND METHODS Twenty-four participants were enrolled and randomly given intradermal injections of BTX and normal saline in both cheeks. Clinician Erythema Assessment (CEA) score, Global Aesthetic Improvement Scale (GAIS) score, skin hydration, transepidermal water loss (TEWL), melanin content, erythema index, elasticity, and sebum secretions were evaluated at baseline and 2, 4, 8, and 12 weeks.

RESULTS On the BTX-treated side, the CEA score significantly decreased and the GAIS score significantly increased. The erythema index decreased at Weeks 4 and 8. Skin elasticity was improved at Weeks 2 and 4 and skin hydration, at Weeks 2, 4, and 8. However, TEWL and sebum secretion did not show significant differences.

CONCLUSION Intradermal BTX injections reduced erythema and rejuvenated the skin effectively and safely in patients with rosacea.

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Since botulinum toxin (BTX) was first approved in 2002 by the US FDA for the treatment of glabellar lines, it has become a standard noninvasive treatment for rhytides and has been used for cosmetic purposes for more than a decade. Many recent studies have examined various indications for the use of BTX, and a therapeutic potential use of BTX in the dermatological field has been suggested.

Several reports have shown the efficacy of intradermal BTX injections for the treatment of facial erythema and flushing. Yuraitis and Jacob[1] first reported the

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possible use of intradermal BTX as an effective treatment for facial erythema. Since then, case reports[2,3] and clinical trials[4,5] have assessed the efficacy of intradermal BTX injections. Other studies have reported the effects of intradermal BTX injections for facial lifting and skin rejuvenation.[6–10] Furthermore, studies have also investigated the ability of BTX to reduce sebum secretion.[11,12] However, no studies published to date have assessed the effects of intradermal BTX injections using quantitative measurements designed for split-face studies.

The authors investigated changes of skin physiology using a noninvasive instrument (a combination corneometer/mexameter/reviscometer/sebumeter device, also known as MPA5) to assess the effectiveness of intradermal BTX injections objectively with quantitative data. The validity and reliability of this instrument have been studied several times in previous studies.[13–15] It is a device that has been used to assess various skin conditions and to evaluate the effectiveness of treatment or any interventions.[16–23]

This study aimed to prospectively evaluate the efficacy and safety of intradermal BTX for facial erythema and skin rejuvenation in subjects with mild to moderate rosacea.

Materials and Methods

Patients and Study Design

This was a single-center, prospective, randomized, double-blind, split-face clinical study. Participants with mild to moderate erythematotelangiectatic (ETR)–type rosacea and facial erythema on the cheeks were recruited. The patients were diagnosed using the standard guidelines of the National Rosacea Society Expert Committee by a dermatologist. Patients who had neuromuscular underlying diseases (e.g., myasthenia gravis, amyotrophic lateral sclerosis, or Lambert-Eaton myasthenic syndrome); had been taking any oral aminoglycoside agent, benzodiazepine, or muscle relaxants from 4 weeks preceding the study; had undergone any aesthetic procedure or received BTX treatment to the face in the preceding 6 months; had an autoimmune disease such as systemic lupus erythematosus; had facial flushing due to menopause; were pregnant or lactating; or had a history of allergy to the study drug were excluded from this study. Patients were also excluded if they had been treated with topical or oral treatment for other facial skin diseases, including rosacea, in the previous 4 weeks. Using computer-generated randomization, each cheek was randomly assigned to receive intradermal BTX or normal saline (NS) injections. A topical anesthetic (EMLA cream; Astra, Westborough, MA) was applied for 30 minutes before treatment and then completely removed. The dilution dose was determined by reviewing previous case reports and clinical trials, considering both efficacy and safety.[1–3,5,6,10–12,24,25] Botulinum toxin (prabotulinumtoxinA; NABOTA, Daewoong Pharmaceutical, Seoul, Korea) was diluted with injectable NS to a concentration of 1 U per 0.1 mL and injected through a 30-gauge, 0.5-mL needle. A total of 15 U of BTX was injected intradermally into one cheek, whereas the other cheek was injected with the placebo (NS). Injections were made at 1-cm intervals for a volume of 0.05 mL at every point (30 points) on an area 5 cm wide and 6 cm long on the cheek in a grid pattern (Figure 1).

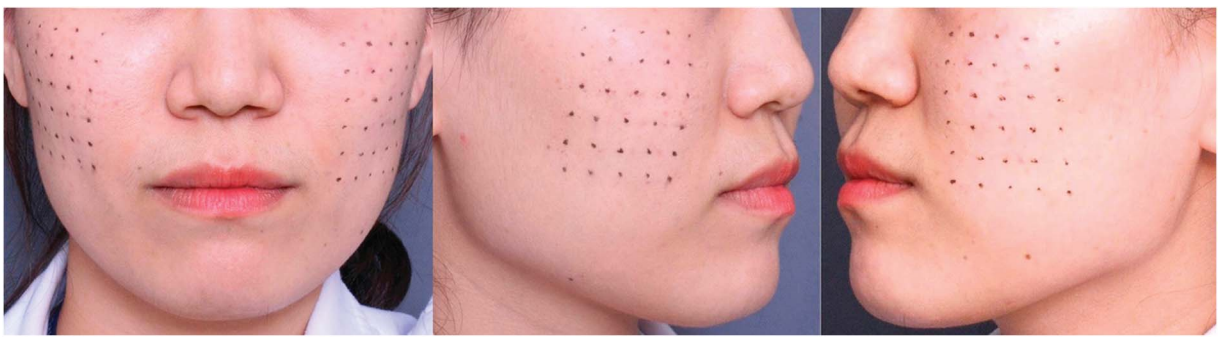


Figure 1. Typical markings at 1 cm^2 with a grid pattern for intradermal injections.

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TABLE 1. Patients' Demographics and Baseline Clinical Characteristics

	BTX-Treated Side (n = *23*)	*Placebo Side* (n = *23*)	p
Age			
Mean ± SD (yrs)	35.2 ± 10.9		
Range (yrs)	21–49		
CEA (mean ± SD)	1.5	1.5	1.00
Hydration (mean ± SD)	51.44 ± 9.75	53.48 ± 10.39	.13
TEWL (mean ± SD)	17.28 ± 5.05	18.77 ± 5.67	.88
Melanin index (mean ± SD)	139.70 ± 35.89	141.84 ± 33.69	.56
Erythema index (mean ± SD)	364.83 ± 83.70	356.68 ± 87.34	.23
Elasticity (mean ± SD)	88.16 ± 90.56	84.38 ± 98.81	.98
Sebum secretion (mean ± SD)	42.00 ± 16.40	43.48 ± 16.88	.05

BTX, botulinum toxin; CEA, Clinician Erythema Assessment; TEWL, transepidermal water loss.

Assessments

The patients underwent an acclimation period of at least 20 minutes to distinguish nontransient erythema from facial flushing. The investigator determined the presence of facial flushing and patient-reported symptoms. If facial flushing was present, photographs were taken after the symptoms disappeared, by using a fixed wooden frame in the same place.

The severity of erythema on both cheeks was assessed using the Clinician Erythema Assessment (CEA) scale, and the Global Aesthetic Improvement Scale (GAIS) was used as previously reported.[26,27] Each patient was evaluated by a nontreating dermatologist at each visit.

Biophysical measurements were performed before treatment and then at 2, 4, 8, and 12 weeks after treatment. A combination corneometer/mexameter/reviscometer/sebumeter device (Courage & Khazaka, Cologne, Germany) was used to measure the stratum corneum hydration level, transepidermal water loss (TEWL), skin melanin and erythema indexes, skin elasticity, and sebum secretions. Each parameter was measured 3 times, and the mean value was used in the analysis.

Safety Assessments

Safety measures included patients' vital signs and adverse events as observed by the investigator or reported by the subject at each visit. Laboratory tests (complete blood count and blood chemistry) were performed at baseline and Week 12.

Statistical Methods

Changes in each of the biophysical parameters were calculated as the pretreatment value minus the post-treatment value for each visit and used in the statistical analyses to minimize the intraindividual variation between the 2 cheeks. To make valuable statements on the effect of the treatment, the results were calculated using the following equation: changes in skin properties (%) = $[(Pt - P_0)/P_0] \times 100$, with Pt being the mean of the measured values of the BTX- or NS-treated side after treatment each week and P_0 being the mean of the measured values of the BTX- or NS-treated side before treatment at baseline. The paired nonparametric Wilcoxon test was used to compare the week-by-week changes in the parameters of the right and left cheeks (i.e., BTX vs placebo). The analyses were performed using the SPSS version 17.0 software for Windows (SPSS, Chicago, IL). The cutoff for statistical significance was set at *p*-values of <.05.

Results

Patients

In total, 24 patients were enrolled. One patient was discontinued from the study because of a protocol violation; at the last visit, the patient had been taking a muscle relaxant for an ankle sprain. Overall, 23 patients completed the study. Most of them were female patients (6 men, and 17 women), and the mean patient age was 35.2 ± 10.9 years. The CEA scores, hydration level, TEWL, melanin index, erythema

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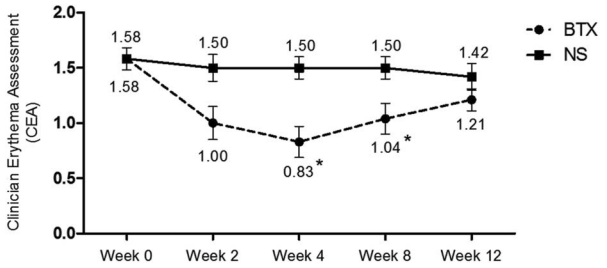


Figure 2. Clinician Erythema Assessment (CEA) score at the follow-up visit. The CEA score significantly decreased in the botulinum toxin (BTX)-treated side at Weeks 4 and 8 ($p < .01$, respectively).

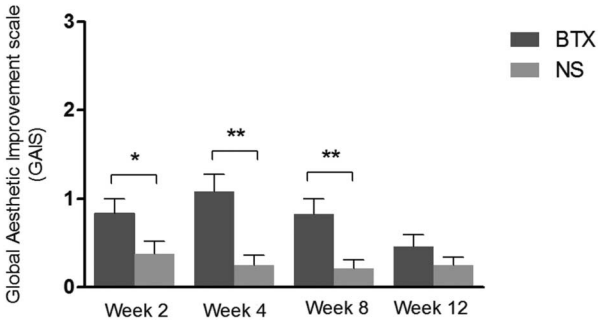


Figure 3. The Global Aesthetic Improvement Scale (GAIS) score was significantly higher on the BTX-treated side at Weeks 2, 4, and 8 ($p < .05$, <.01, and .01, respectively). BTX, botulinum toxin.

index, elasticity, and sebum secretions did not differ between the treated and placebo sides (Table 1).

Safety

None of the enrolled subjects experienced significant adverse effects such as an allergic reaction, facial palsy, or severe paralysis of the muscles adjacent to the point of injection, during or after the study. All the patients experienced mild erythema after the injections (lasting a mean of 2.21 days). The pain during the injections was tolerable in all the subjects and comparable between the sides.

Improvement Score

The mean CEA score of the BTX-treated side was significantly lower at Weeks 4 and 8 ($p < .01$; Figure 2). The mean GAIS scores are shown in Figure 3. The mean GAIS scores of the BTX-treated side were significantly high at Weeks 2, 4, and 8 ($p < .05$, <.01, and <.01, respectively). The photographs taken of 1 patient are shown in Figure 4.

Biophysical Assessments

The erythema index was decreased in the BTX-treated side throughout the 12-week study period, and statistical differences were observed at Weeks 4 and 8 as compared with the baseline values ($p < .01$ and <.01, respectively). A significant difference in the change in the erythema index was observed between the BTX- and NS-treated sides at Weeks 4 and 8 ($p < .05$ and <.05, respectively). The calculated percentage change of the erythema index also showed differences, which were significant at Weeks 4 and 8 ($p < .05$ and <.01, respectively). However, the melanin index did not differ significantly between sides during the study period (Table 2).

The authors used a reviscometer to measure skin elasticity, with lower reviscometer values indicating higher elasticity. On the BTX-treated side, the reviscometer values were significantly decreased at Weeks 2 and 4 as compared with the baseline value ($p < .01$ and <.05, respectively). At Weeks 2 and 4, the changes in the reviscometer values on the BTX-treated side were significantly greater than those of the NS-treated side ($p < .01$, respectively; Table 2). The calculated changes in the reviscometer values (%) on the BTX-treated side were statistically significant as compared with the NS-treated side ($p < .05$ and <.01, respectively; Table 2).

A corneometer was used to measure hydration levels of the stratum corneum, with higher corneometer values reflecting higher hydration levels. On the BTX-treated side, the hydration level was improved throughout the 12-week study period, and significant improvements were observed at Weeks 2, 4, and 8 when compared with the baseline values ($p < .01$). At Week 2, improvement of hydration was also observed on the NS-treated side ($p < .05$). The changes in corneometer values were higher on the BTX-treated side than on the NS-treated side at Weeks 2, 4, and 8 ($p < .05$, <.01, and <.01 respectively; Table 2). The calculated change in corneometer values (%) was also higher on the BTX-treated side at Weeks 2, 4, and 8 ($p < .05$, <.01, and <.01, respectively; Table 2).

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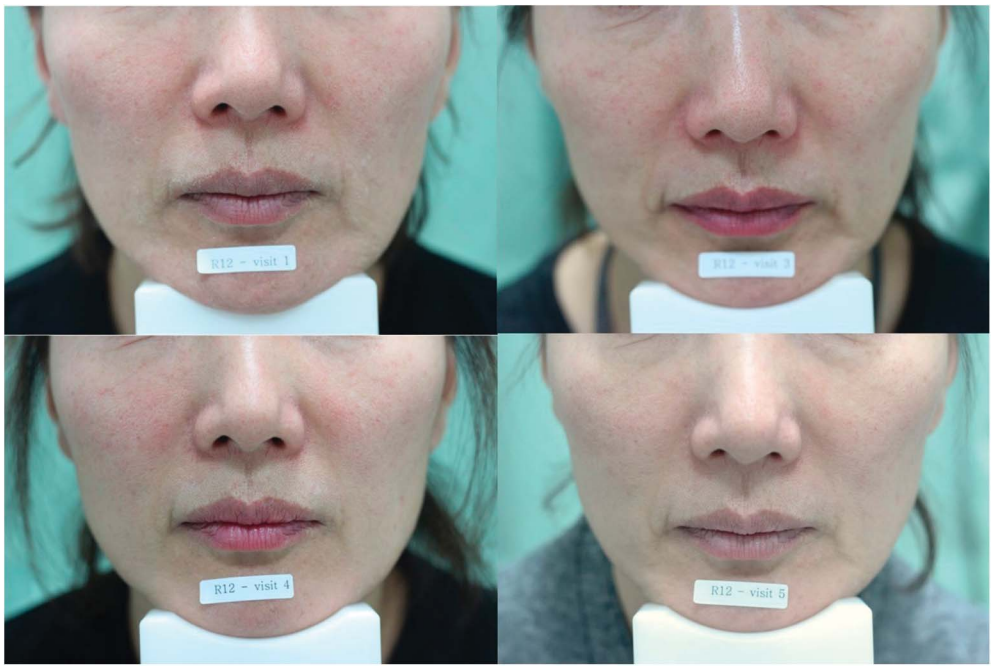


Figure 4. Clinical response in a 48-year-old woman after intradermal botulinum toxin (BTX) injections (left cheek) and normal saline injections (right cheek). Erythema and clinical improvement can be observed on the BTX-treated side. (Top row, left to right: at baseline and Week 4; bottom row, left to right: at Weeks 8 and 12.)

The authors measured the TEWL of both cheeks, with lower TEWL values reflecting a healthy skin condition. The TEWL did not change significantly during the study period, and changes in the TEWL did not significantly differ between the BTX- and NS-treated sides throughout the study period ($p > .05$; Table 2).

The sebum secretion did not differ significantly throughout the 12-week study period, and no statistically significant differences were found between the BTX- and NS-treated sides ($p > .05$; Table 2).

Discussion

This split-face study evaluated the efficacy and safety of intradermal BTX injections using biophysical measurements, and the authors' findings indicate that it is a favorable treatment for facial erythema and rejuvenates the skin in patients with rosacea.

The CEA score and erythema index on the BTX-injected side were significantly decreased at Weeks 4 and 8. Previous studies demonstrated that intradermal injection could improve facial flushing and erythema.[3,5] However, in previous studies, only the evaluator's score was used to assess the effect, but this study objectively showed the efficacy of intradermal BTX injections quantitatively by measuring the erythema index. The vascular endothelial growth factor (VEGF) is known to be the major angiogenetic factor that plays a role in the development of nontransient erythema in subjects with rosacea. A previous study demonstrated that BTX could suppress VEGF expression through inflammatory modulation by interleukin-8 downregulation in subjects with chronic cystitis.[28,29] The authors presumed that the anti-VEGF effect of BTX injection could affect the reduction of CEA and erythema index in this study. However, on the contrary, BTX could affect vasodilation and VEGF expression. Previous studies demonstrated that BTX could affect the cutaneous vasculature by increasing VEGF at the molecular level in the skin flap model.[30–34] The mechanism of how BTX affects the expression of VEGF is unknown; hence, the authors could speculate that the condition of the injected site and injection dose might have caused these differences in the results. Further molecular studies will be needed in the future to determine the relationship between BTX and VEGF. Several previous studies showed that BTX injections could reduce facial flushing by

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TABLE 2. Differences in Hydration Level, Melanin Index, Erythema Index, and Skin Elasticity Calculated on a Week-By-Week Basis

	Mean Difference			*Calculated Percent Changes (%)*		
	BTX-Treated Side	*NS-Treated Side*	p	*BTX-Treated Side*	*NS-Treated Side*	p
Erythema index						
Week 2	−7.00 (−22.4, 14.0)	−2.00 (−21.5, 15.4)	.54	−2.36 ± 6.79	−0.08 ± 11.9	.52
Week 4	−19.7 (−47.3, 8.00)	−0.63 (−27.0, 19.7)	<.05	−6.02 ± 7.70	0.52 ± 13.4	<.05
Week 8	−19.3 (−46.6, −2.67)	2.97 (−34.7, 17.3)	<.05	−5.17 ± 10.4	1.27 ± 14.2	<.01
Week 12	−17.7 (−45.4, 29.3)	3.03 (−36.8, 14.0)	.07	−4.87 ± 12.4	0.20 ± 12.9	.06
Melanin index						
Week 2	−1.40 (−9.70, 13.0)	1.35 (−9.30, 7.30)	.92	5.94 ± 31.9	0.96 ± 10.7	.63
Week 4	−0.40 (−15.0, 7.60)	−3.60 (−8.00, 4.70)	.38	−2.58 ± 23.8	−3.22 ± 12.4	.83
Week 8	−0.33 (−13.3, 4.67)	−5.97 (−14.0, 6.20)	.84	15.5 ± 106	2.07 ± 23.6	.47
Week 12	2.30 (−26.0, 10.7)	−3.03 (−30.7, 10.3)	.33	6.02 ± 6.93	−2.12 ± 18.3	.86
Elasticity						
Week 2	−16.9 (−29.6, 9.80)	2.26 (−12.3, 7.80)	<.01	−15.6 ± 28.5	3.01 ± 29.8	<.05
Week 4	−7.75 (−31.1, 4.00)	−1.13 (−9.84, 4.51)	<.01	−17.9 ± 31.1	−1.29 ± 25.4	<.01
Week 8	−2.14 (−33.2, 7.45)	0.68 (−21.9, 10.0)	.11	−4.69 ± 40.9	6.65 ± 36.9	.14
Week 12	2.05 (−15.3, 17.9)	3.46 (−18.5, 16.5)	.90	9.21 ± 47.0	10.9 ± 34.0	.83
Hydration						
Week 2	6.30 (2.60, 11.5)	0.47 (−2.45, 5.76)	<.05	12.9 ± 12.9	4.00 ± 14.9	<.05
Week 4	6.83 (4.33, 10.9)	0.07 (−1.89, 4.33)	<.01	16.9 ± 14.2	3.60 ± 11.4	<.01
Week 8	9.13 (4.80, 13.7)	1.40 (−0.02, 3.23)	<.01	17.2 ± 14.4	4.71 ± 8.57	<.01
Week 12	2.27 (−4.37, 8.90)	1.63 (−3.70, 6.03)	.41	7.26 ± 21.2	2.59 ± 16.8	.32
TEWL						
Week 2	−0.50 (−7.60, 2.80)	0.50 (−3.40, 3.40)	.18	−3.39 ± 2.94	−7.7 ± 4.78	.98
Week 4	0.10 (−3.70, 4.30)	1.40 (−2.30, 5.10)	.36	1.96 ± 3.26	9.90 ± 2.42	.38
Week 8	1.80 (−1.10, 5.60)	1.40 (−2.20, 5.50)	.81	1.60 ± 4.08	2.16 ± 4.62	.93
Week 12	1.00 (−1.90, 6.60)	0.40 (−1.20, 3.90)	.67	1.66 ± 3.55	1.04 ± 2.55	.70
Sebum secretion						
Week 2	1 (−4, 5)	1 (−5, 3)	.73	1.10 ± 1.47	−0.65 ± 1.45	.72
Week 4	1 (−4, 5)	−4 (−6, 5)	.35	2.71 ± 2.16	−0.47 ± 1.94	.25
Week 8	−1 (−7, 5)	−3 (−4, 2)	.46	0.43 ± 1.95	−1.77 ± 1.52	.64
Week 12	−2 (−5, 3)	−2 (−4, 1)	.72	1.73 ± 1.98	−3.39 ± 1.37	.53

BTX, botulinum toxin; NS, normal saline; TEWL, transepidermal water loss.

blocking acetylcholine release from the autonomic peripheral nerves, which directly inhibits vasodilation and neurotransmitters (e.g., vasoactive intestinal peptide, pituitary adenylyl cyclase–activating polypeptide, and/or nitric oxide). The authors presumed that the reduction in facial flushing also affects the status of chronic inflammation of the cutaneous vasculature.

Elasticity was significantly improved at Weeks 2 and 4 after the injections as compared with the NS-injected side. The intramuscular BTX injection is already known to achieve wrinkle correction by muscular relaxation within 2 weeks after injection, although the effects gradually diminish 2 to 3 months later.[10] In this study, the authors observed that the elasticity improved at Week 2 and peaked at Week 4. They assume that low-concentrated intradermal BTX injections cause changes in the facial expression muscles that the subjects could not recognize but may smooth out fine wrinkles and improve skin elasticity in a shorter duration.[35,36]

In this study, the hydration values of the stratum corneum were significantly improved at Weeks 2, 4, and 8; however, TEWL did not improve after

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treatment. The authors speculate that the early rejuvenation effect of intradermal BTX injections is related to the lymphatic accumulation as a result of mild paresis of the facial muscles.[36] Moreover, the late rejuvenation effect that gradually appeared after Week 4 was related to collagen neogenesis. Previous studies found that BTX could induce collagen production and downregulate collagen degradation in in vitro and in vivo studies.[6,35] The authors also assume that the minimal trauma due to multiple intradermal needle injections also impacts skin rejuvenation. The minimal trauma induces a biological response in the lymphatic and immune systems, which affects the physiological alterations of the skin.[36–39]

No significant improvement in sebum secretion was observed in this study. By contrast, Rose and Goldberg[12] reported that intradermal BTX significantly reduced sebum production in the forehead region. However, the results require careful interpretation. Because the forehead tends to be an oilier region than the cheek, the treatment to the cheek region in the present study could not induce significant improvements in sebum secretion. Additional studies are needed to determine whether it reduces sebum secretion depending on the intradermal BTX injection site.

This study was subject to some limitations. The number of enrolled patients was small, and the treatment follow-up period was relatively short. Further large-scale long-term follow-up studies are needed to reach a confirmative conclusion. In addition, further research is needed to reach consensus for dilution, doses, injection sites, and detailed injection methods. Several studies are underway, but results are currently insufficient to share the approved protocol. In the present study, all the subjects were injected with the same dose (total of 15 U of BTX) regardless of their characteristics such as facial erythema severity. Further clinical trials are needed to elucidate the effect of intradermal BTX injections and reach consensus about the protocol.

In conclusion, the authors evaluated the efficacy and safety of intradermal BTX injections in patients with facial rosacea. These data showed that the intradermal BTX injection would be useful for decreasing facial erythema and improving other skin biophysical characteristics. Erythema, skin elasticity, and skin hydration were improved with intradermal BTX injections. No reduction in sebum secretion was observed in this study. However, these results are not a direct refutation of previous research results and require careful interpretation because of differences in the injection area. Despite the small number of subjects, the strength of this study is that it was conducted as a split-face study with objective evaluation using CEA and GAIS scores and quantitative measurement. These findings suggest that intradermal BTX injections may be a useful and tolerable treatment option for facial erythema and skin rejuvenation in subjects with ETR-type rosacea.

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